SPE-228928-MS

Enhancing Reservoir Connectivity and Production Via Novel Multilateral Drilling Techniques

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Maximizing reservoir contact in heterogeneous, low-connectivity carbonates remains challenging where conventional stimulation (matrix acidizing, acid fracturing, hydraulic fracturing) underperforms due to unfavorable lithofacies, thin pay, complex diagenetic overprints, and water-thief connectivity. This paper documents the field application of hyper-short-radius Mechanical Radial Drilling (MRD) to create large-diameter (≈2-3/4 in) lateral channels of ~45 ft each, placed orthogonally at a single TVD from an existing horizontal wellbore in a mature UAE carbonate reservoir. The campaign used a retrievable whipstock, solid-expandable anchor with orientation shoe, a double-bend slim mud motor capable of ~200–215°/100 ft dogleg, and a small-diameter PDC bit; the design enables high-density laterals with re-entry capability for diagnostics and subsequent treatments [1]. Post-intervention, the well initially increased from ~200 to ~400 BOPD, followed by production decline attributable to high near-wellbore skin (~ +30) diagnosed via pressure-transient analysis (PTA). A targeted bullhead acid cleanup to remove filter cake restored connectivity, delivering ~3× improvement in PI with stable water cut over >6 months, fully recovering operational costs. The paper details candidate screening based on petrophysical and geological considerations, operational execution (including Logging While Tripping, LWT), results, lessons learned, and an economic evaluation framework for scaling MRD in similar reservoirs [2–6].

Introduction

Mature carbonate reservoirs in the Middle East commonly exhibit strong heterogeneity and compartmentalization because of facies architecture and diagenetic overprints (e.g., stratiform/selective dolomitization), often resulting in variable permeability and limited lateral/vertical communication. In such systems, standard stimulation may have limited or short-lived impact, especially where acidizing fails to create sustained conductive pathways or where hydraulic fracturing risks connecting with aquifers [7,8].

Mechanical Radial Drilling (MRD) offers an architecture-driven approach to improve near-wellbore and mid-field connectivity by extending short laterals from an existing wellbore into undrained compartments. Unlike Radial Jet Drilling (RJD), which uses high-pressure jetting and can struggle in hard formations, Mechanical Radial Drilling (MRD) uses a slim mud motor and PDC bit to drill laterals with controlled azimuth/trajectory, enabling survey verification and re-entry for future diagnostics or stimulation [9–11]. MRD implementation enables 2-3/4 in OD channels up to ~13.71 m (≈45 ft) long, with dogleg severities up to ~215°/100 ft, retrievable whipstock guidance, and solid-expandable anchor/orientation shoes that allow precise placement and subsequent re-entry [1].

Well Objectives

The central aim of this well was to assess both the technical viability and operational dependability of advanced stimulation technology for mechanical radial drilling, directionally controlled channels within the geological boundaries of the field. The initiative specifically targeted the creation of four radial channels, each designed to be approximately 45 feet in length and strategically positioned at distinct measured depth intervals (Figure. 1), thereby enabling access to previously untapped reservoir segments situated beneath the mother bore [9,11,15].

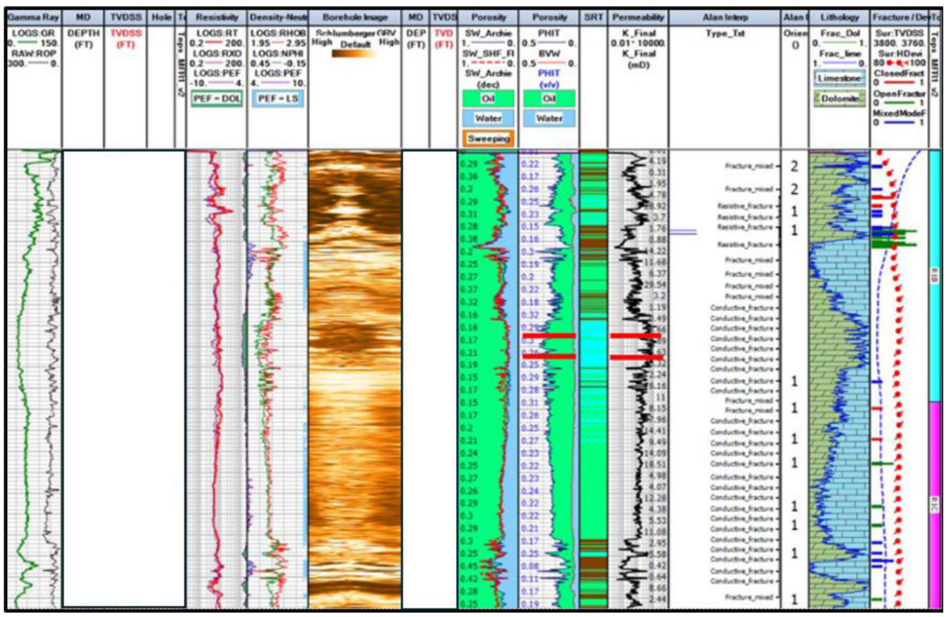

Figure 1—Open Hole Interpretation drilling paths selected based on rock properties.

Upon successful channel construction, the plan called for acid stimulation to further improve connectivity and optimize reservoir productivity.

Candidate Selection Criteria

The selection of suitable wells for Mechanical Radial Drilling deployment is critical to maximizing the technology's impact on reservoir performance. In this case study, the candidate reservoir is located in the UAE and is characterized by complex geological features (Figure. 2), including intercalation of hard and soft layers due the diagenetic print on the reservoir, resulting in variable permeability and poor natural connectivity [9].

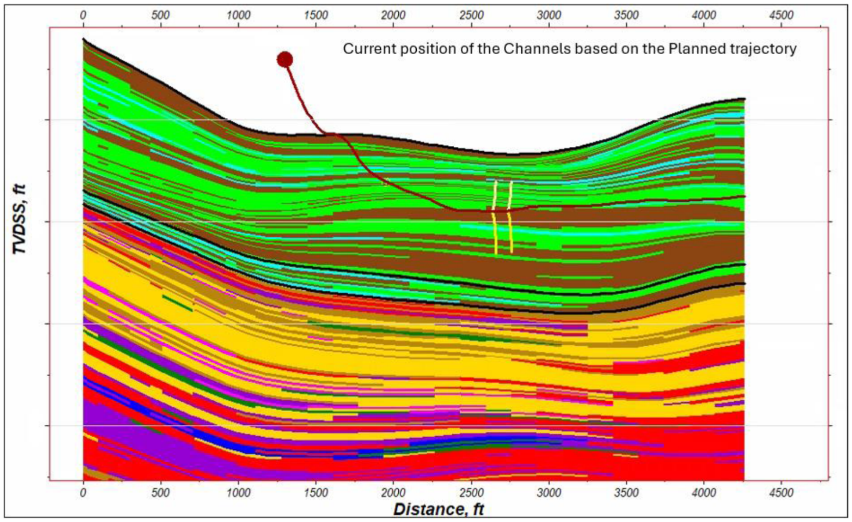

Figure 2—Reservoir Model (Dolomitic facies in Light Green)

The reservoir exhibits complex geological and diagenetic features that significantly influence its performance. The reservoir is characterized by a variable porosity, ranging from 8.6% to 26.9%, with permeability going from 0.1 mD reaching up to 1000 mD. Diagenetic processes, particularly dolomitization and fracturing, have enhanced reservoir quality by creating secondary porosity and improving fluid flow pathways. These features are especially, where dolomite-rich zones exhibit superior connectivity and reservoir performance. However, there are areas where lithological barriers prevent vertical and horizontal communication across units, confirmed by production data and simulation models. Overall, the field geological complexity and diagenetic evolution play a pivotal role in its fluid distribution, pressure behavior, and recovery efficiency [9,15].

These conditions have historically limited the effectiveness of conventional stimulation techniques (dolomites have limited reaction to acid and hydraulic fracturing stimulates the natural features (fractures, dissolution features, perm streaks among others) of the reservoir that connect with water channels) and contributed with the poor productivity of the wells in this area.

Initial screening focused on wells exhibiting the following characteristics:

- **Not improvement of PI** despite stimulation efforts
- **Permeability heterogeneity**, ranging from 0.1 to 1000 millidarcies (mD)
- **Thin pay zones** with poor vertical communication
- **Low sustained water cut**, to avoid stimulating water channels

The selected well had initially produced up to 700 barrels of oil per day (BOPD) but had declined to approximately 200 BOPD over six years due to reservoir depletion. Previous acid stimulation and matrix treatments yielded only short-lived improvements, indicating that the root issue was limited reservoir contact rather than formation damage.

Reservoir modeling and dynamic simulation were used to identify undrained zones and optimal lateral placement. The well's structural integrity, completion design, and available access points were also evaluated to ensure compatibility with the Mechanical Radial Drilling system. The final candidate was chosen based on its potential for incremental recovery, operational feasibility, and alignment with the technology's strengths in enhancing connectivity in low-permeability, layered formations.

Job Execution and Operational Workflow

The successful deployment of the Mechanical Radial Drilling technology in the selected well required meticulous planning, precise execution, and real-time monitoring to ensure optimal reservoir contact and mechanical integrity. The operation was divided into several key phases: pre-job modeling, equipment mobilization, lateral drilling, and post-job evaluation.

Pre-Job Planning

Reservoir simulation, geological modeling and open hole logging interpretation were used to identify target zones for lateral placement. The goal was to intersect undrained compartments while avoiding water-bearing layers as portraited in (Figure. 3). A whipstock system (Figure. 4) was designed to guide the drilling assembly into four orthogonally oriented trajectories from the main horizontal borehole [9–11,1].

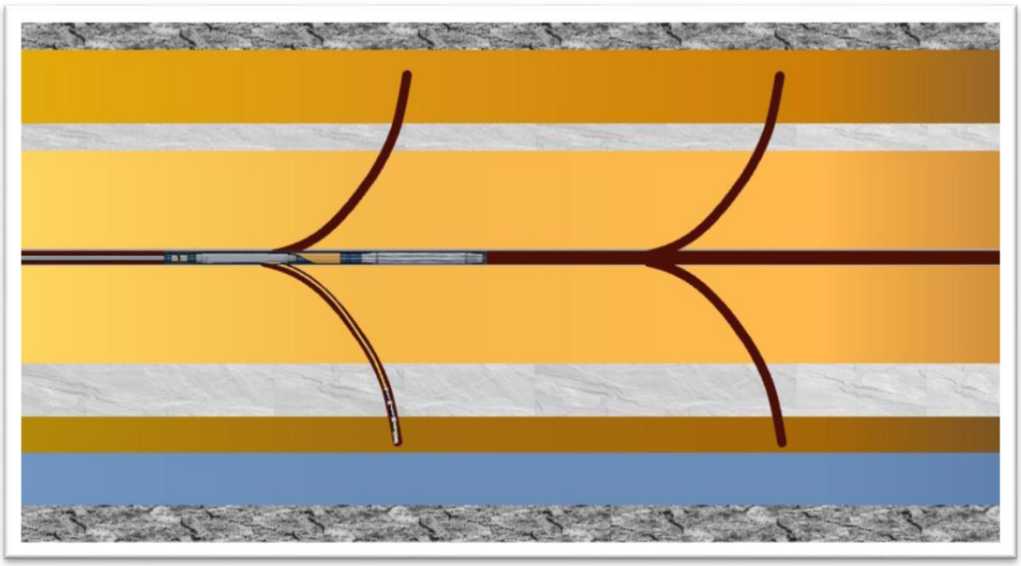

Figure 3—Configuration of channels during MRD technology

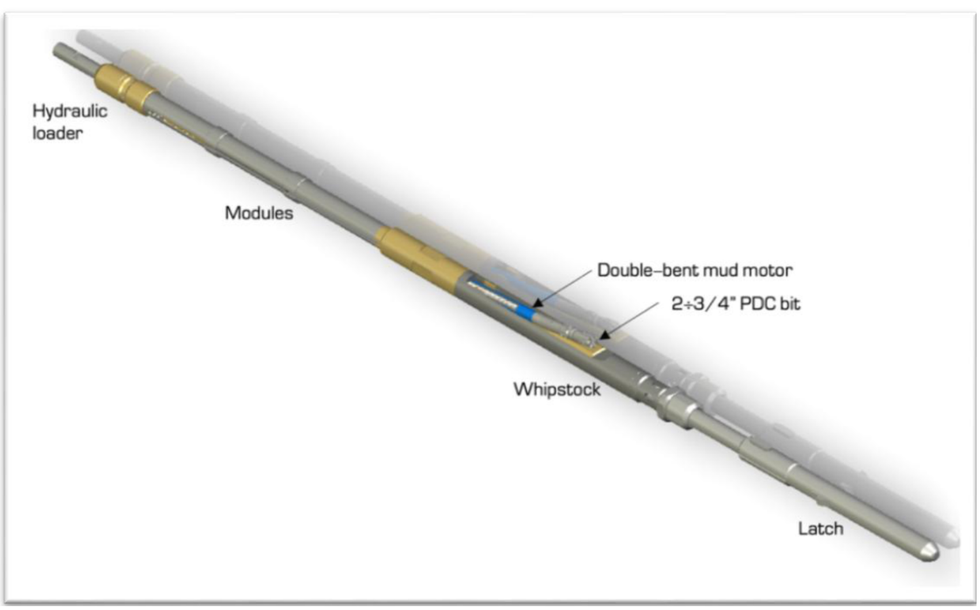

Figure 4—BHA for Mechanical Drilling

Equipment and Tools [1]

The Mechanical Radial Drilling system utilized (Figure. 5):

- A **retrievable whipstock** for directional control
- A **double-bend mud motor** capable of achieving dogleg severities up to 215°/100 ft
- A **2-3/4 inch PDC bit** optimized for high ROP across mixed lithologies

Figure 5—Whipstock, retrievable anchor and mud motor

An **expandable anchor** with an integral orientation shoe, to stabilize the assembly in both open and cased hole environments and provide the directional reference for all drilled channels.

Logging While Tripping

A Logging While Tripping (LWT) operation was executed to overcome production logging limitations caused by operational constraints and tool deployment challenges. This approach enabled data acquisition during the condition trip of the open hole section, optimizing rig time and reducing intervention costs. The logging suite included gamma ray, resistivity, density–neutron, and caliper tools, providing a comprehensive formation and borehole evaluation. Density and neutron logs were used to estimate porosity, while resistivity measurements supported fluid saturation analysis. The caliper tool assessed borehole geometry, identifying washouts, tight spots, and irregularities essential for ensuring wellbore integrity and proper anchor placement. These measurements validated oil-unswept intervals and confirmed selections from dynamic and static models. The integration of these datasets allowed for a robust understanding of the borehole's actual condition and geometry, supporting the intervention plan [2,3].

Drilling Procedure

The drilling process commenced with a thorough assessment of the channels depths using caliper measurements. This step was vital for mapping the borehole and identifying potential irregularities in diameter. Simultaneously, differential sticking risk events were evaluated to ensure safe anchor positioning at the required depth. These assessments helped mitigate the risk of drill string becoming immobilized due to pressure differences or debris. The anchor was carefully positioned based on these preliminary measurements and risk analyses, providing a stable foundation for lateral channels drilling [1,2,3].

Each lateral channel was meticulously drilled to an approximate length of 45 feet, ensuring that the designed subsurface architecture was precisely achieved. The drilling procedure required that the integral whipstock's orientation was accurately set to the desired azimuth relative to the downhole anchor orientation shoe. The channel orientation setting is performed on surface while making up the running tool and BHA. The drilling BHA is contained inside the running tool and is only released after the anchor has been engaged and the whipstock toolface has been locked in position facing the desired azimuth. The drilling BHA drills a predictable fixed trajectory determined by the specific design geometry of the tools. After each run, the integral whipstock was re-oriented at surface for the next channel. This allowed for adjustments in the drilling trajectory and ensured that subsequent channels were oriented as intended. The goal was to achieve an orthogonal orientation for each channel, thus maximizing spatial separation. Continuous parameter control and monitoring were conducted throughout drilling, allowing operators to respond quickly to any changes in hole conditions or tool performance. Upon completion of channels #1 and #4, a detailed survey was conducted (Figure. 7). This survey verified the anchor-set and trajectory orientation, providing assurance that the channel placement matched engineering specifications. The channels were planned to deliver between 150 deg/100 ft and 200 deg/100ft dogleg severity. The 2 channels surveyed demonstrated that the doglegs achieved (average >150 deg/100ft) comfortably fell inside the target window and achieved sufficient TVD (20-23 ft) to connect the targeted formation intervals above and below the motherbore [12].

Specialized drilling fluids were formulated and tailored to the unique conditions of the borehole, The properties of the drilling fluids such as viscosity, density, and chemical composition were carefully controlled to maintain borehole stability and minimize the risk of collapse or fracturing. Differential sticking risk was reassessed at key stages, ensuring that any new risks were identified and addressed promptly. Safety protocols were enforced throughout the operation, with procedures designed to protect personnel, equipment, and the well structure.

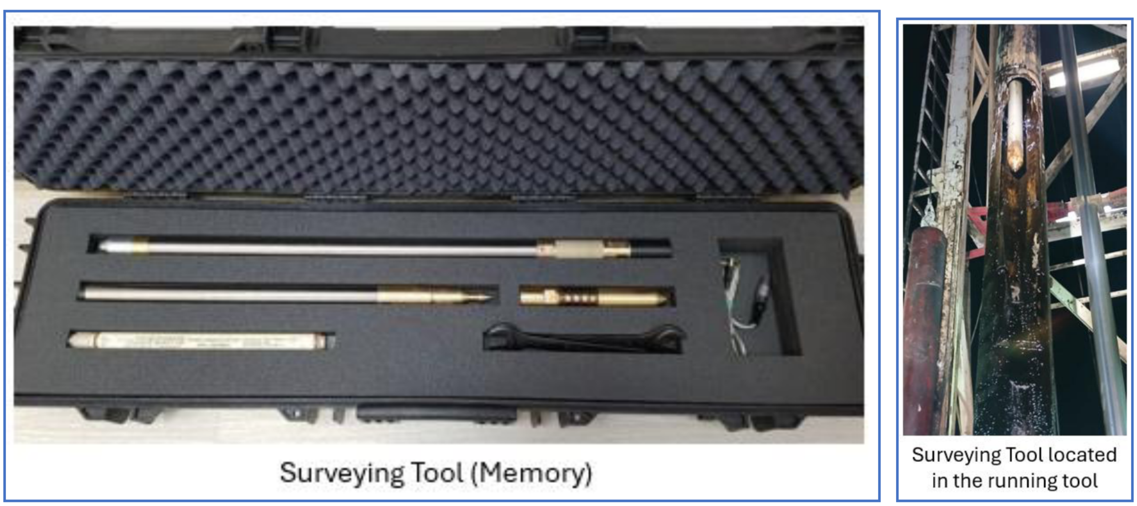

Figure 6—Survey Tools

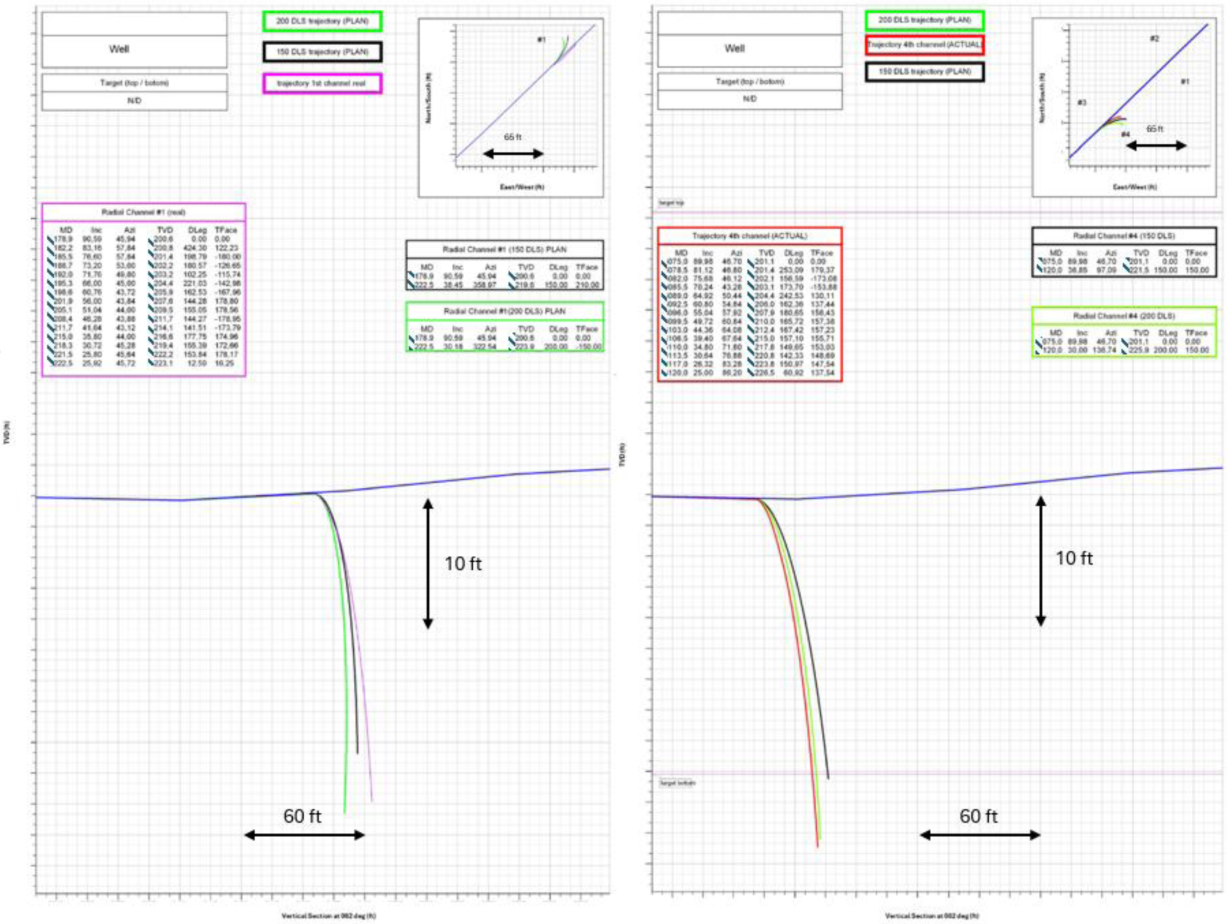

Figure 7—Surveyed Trajectories

Operational Challenges

Key challenges included:

- Navigating through interbedded formations with variable hardness
- Confirming trajectory and channel placement in high dogleg environments
- Ensuring mechanical integrity of small-diameter laterals
- Avoiding premature water breakthrough

These were mitigated through adaptive drilling parameters, close operational monitoring, and contingency planning.

The depth selection of the channels are based on:

- Best reservoir properties seen along the horizontal section in the log analysis (Open hole data) and the properties near the borehole estimated in the static model.
- Seismic interpretation was used to map potential fracture corridors and avoid them when choosing the channels placement.
- The direction of the channels is at 45 / 135 / 225 and 315 deg of azimuth respectively. This way we can maximize the contact of the channels with the upper and lower layers. Figure.7
- Logs will be obtained during the condition trip (Logging While Tripping) to check the status of the borehole, saturation, porosity and permeability.

Production Results and Performance Evaluation

Following the successful execution of the Mechanical Radial Drilling intervention, the well was brought back online with an initial production rate of approximately 400 barrels of oil per day (BOPD). However, production rapidly declined and became unsustainable. Pressure transient analysis (PTA) revealed a high skin factor of around 30, attributed to filter cake development due to the viscosified fluid drilling and the dynamic losses experienced while drilling the channels. To address this, coiled tubing acid treatment was performed to dissolve the filter cake, resulting in a threefold increase in productivity (Figure. 8). The well-maintained stable production for over six months post-treatment, effectively recovering the operational costs of the intervention. Additionally, the water cut remained stable, indicating that the laterals successfully targeted oil-rich zones while avoiding areas with high water saturation [4,5,10, 13].

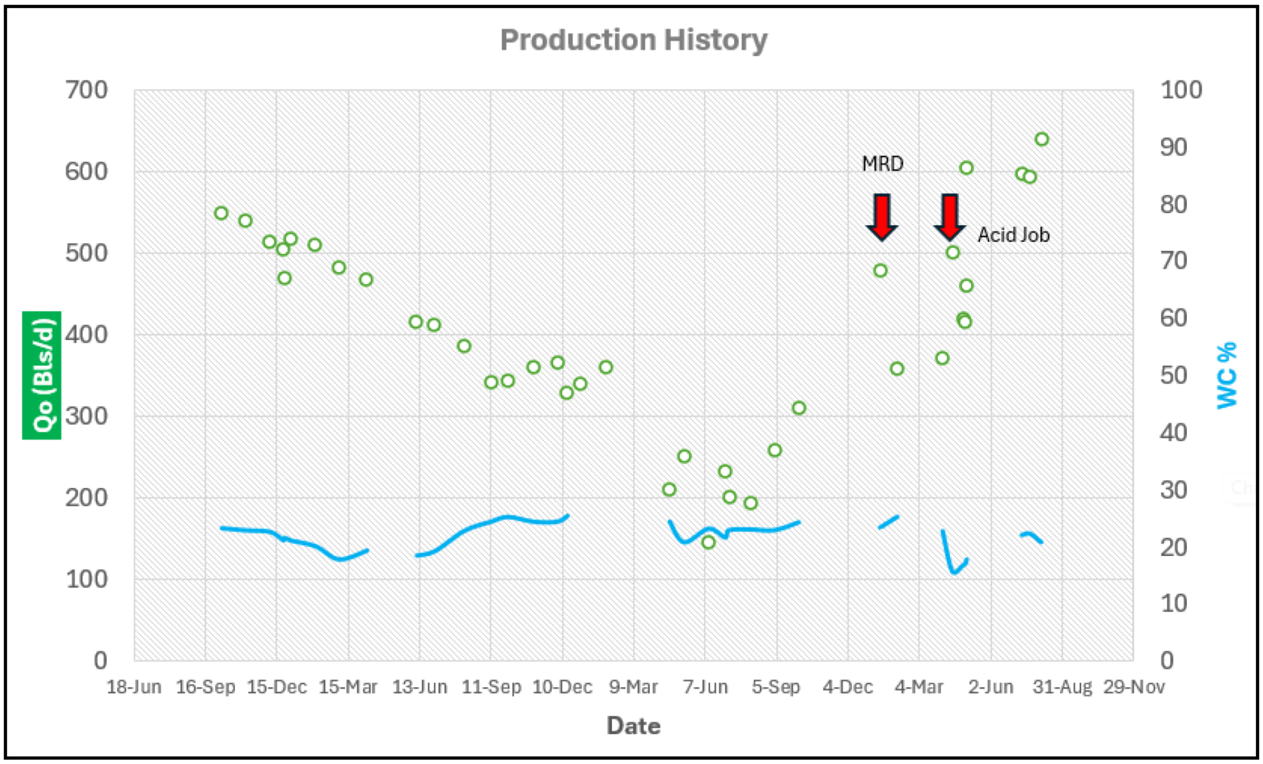

Figure 8—Production Recovery

Economic Impact

The cost of the Mechanical Radial Drilling operation was significantly lower than that of drilling a new multilateral well or performing repeated stimulation treatments. When normalized against incremental production, the **cost-per-barrel of additional oil recovered** was 5 MM NPV, making the approach economically attractive for similar mature or marginal fields [10,14].

Operational Insights

Key lessons learned from the operation include:

- The importance of **accurate reservoir modeling** to guide lateral placement
- The value of **re-accessible laterals** for future stimulation or diagnostics

Overall, the intervention demonstrated that this lateral drilling system can be a powerful tool for revitalizing underperforming wells in complex geological settings.

The major highlights and challenges are summarized below.

- Health, Safety, and Environment (HSE) Performance: Throughout the scope of the operations, rigorous safety protocols ensured that no incidents relating to health, safety, or environmental issues were recorded or attributed to any phase of the project. Proactive risk assessments and a culture of safety compliance contributed to flawless operational execution.
- System Handling in Horizontal Open Hole: The team demonstrated advanced technical expertise by managing the operations within horizontal open hole conditions. Equipment was deployed, operated without any operational setbacks, underlining both the reliability of the technology and the proficiency of the personnel involved.
- Channel Drilling Precision: All channels required by the project were drilled in strict accordance with the planned orientation and approved design specifications. This achievement underscores the meticulous preparation and precision-focused drilling methods that were employed to meet engineering objectives and maintain quality standards.
- Single Anchor, Multiple Channel Drilling: Utilizing a single anchor for the operation, the team efficiently drilled four distinct channels with the aid of a long spacer element incorporated into the Bottom Hole Assembly (BHA). This innovative approach minimized the need for multiple tool deployments, streamlined operations, and maximized time efficiency.
- Anchor Success: The process of setting and retrieving the open hole anchor was executed with accuracy and reliability, with the anchor mechanism performing flawlessly during both deployment and recovery. This operation was key to maintaining integrity and demonstrated the value of coordinated teamwork between surface and downhole crews.
- Channel Entry and Re-Entry: The ability to enter and subsequently re-enter drilled channels was proven through systematic testing and validation.
- Channel Surveying: Comprehensive surveys were conducted for channels 1 and 4, yielding detailed insights into the structural integrity and conformity to design. Survey outcomes provided valuable documentation for validation purposes and will inform strategies for future channel development or enhancements.
- Managing Drilling Challenges: The team effectively confronted and managed complex drilling challenges, including differential sticking and increasing fluid losses to the well. By implementing adaptive fluid management strategies and maintaining vigilant real-time monitoring, the team preserved wellbore stability and ensured uninterrupted drilling progress.
- No Stuck-Pipe Events: The operation was free from any stuck-pipe incidents, a common and potentially costly problem in drilling environments. This outcome was achieved through careful planning, risk mitigation, and the steadfast application of best practices throughout the drilling program.
- Timely Completion: The operational program was concluded within the planned timeframe, illustrating exemplary project management and execution. All phases were coordinated efficiently, from initial setup to final completion, reflecting the dedication, skill, and discipline of the entire operational team.
- Differential Sticking Risk and Management: The operation faced continuous risks of differential sticking, which led to frequent hang-ups and the need for overpulls. This created operational inefficiencies and increased the complexity of the process, as each hang-up required additional actions and monitoring.
- Anchor Shoe Flushing Requirements: After almost every operation, it was necessary to flush the anchor shoe to establish a satisfactory connection with the anchor. This repetitive requirement added to the operational time and indicated persistent challenges in maintaining reliable anchor connections.

- Debris and Filter-Cake Accumulation: Accumulation of filter-cake, low side debris, and cuttings on the latch and Bottom Hole Assembly (BHA) during Run-In-Hole (RIH) posed two significant risks: it prevented effective connection with the anchor and threatened to plug the bit nozzles. These issues could compromise tool functionality and wellbore cleaning, necessitating constant attention.
- Survey Channel Access and Well Stuck Risk: The initial attempt to access the Up-survey channels (channels 2 and #3) was unsuccessful. Due to elevated stuck pipe risk and increasing losses in the well, priority was given to completing the well program successfully. Consequently, the team decided against performing a cleanout run or modifying the survey tool geometry, accepting some limitations in surveying to ensure overall well objectives were met and to minimize further operational risks.

Conclusions

The application of Mechanical Radial Drilling technology in a mature, low-connectivity reservoir in the UAE has demonstrated a significant advancement in well productivity and reservoir management. By deploying large-diameter, orthogonally oriented lateral production channels from an existing horizontal wellbore, the intervention successfully enhanced reservoir contact, improved fluid flow paths, and unlocked previously undrained hydrocarbon zones.

The operation resulted in a threefold increase in oil production while maintaining a stable water cut, validating the technical and economic viability of this approach.

This case study highlights the potential of Mechanical Radial Drilling as a cost-effective alternative to conventional stimulation and multilateral well strategies, particularly in complex, compartmentalized, or thinly layered formations. The ability to re-enter and evaluate each lateral further enhances the long-term value of technology.

In conclusion, Mechanical Radial Drilling represents a paradigm shift from stimulation-centric methods to architecture-driven reservoir enhancement. Its successful deployment offers a scalable solution for revitalizing underperforming assets and maximizing recovery in challenging reservoir environments.

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